PES UNIVERSITY

Department of Computer Science & Engineering

Object Oriented Analysis and Design

UE23CS352B

Assignment 8 Submission

Name of the Student	Pranav Hemanth
SRN	PES1UG23CS433
Section	G
Department	CSE
Campus	RR

Object Oriented Analysis and Design (UE23CS352B) - Lab 8

Student Management System

Department of Computer Science & Engineering
Object Oriented Analysis and Design

UE23CS352B

Problem Statement - Student Management System

1. Problem Statement

The objective of this lab is to build a web-based Student Management System that performs basic CRUD operations. The application is required to implement the Model-View-Controller (MVC) architectural pattern using Spring Boot, Hibernate (JPA), and an H2 relational database.

2. Technologies Used

Backend Framework: Java Spring Boot
Architecture Pattern: Model-View-Controller (MVC)
Persistence: Spring Data JPA / Hibernate
Database: H2 (In-Memory Database)
Frontend: HTML, CSS, Thymeleaf

3. Architecture Overview

The application follows a standard three-tier architecture:

a. Model (Data Layer): The Student entity represents the database schema. It is mapped to the default STUDENT table using JPA annotations.
b. View (Presentation Layer): The HTML/Thymeleaf template (index.html) provides the user interface for registration and viewing records.
c. Controller (Application Logic): The StudentController intercepts incoming HTTP requests (GET, POST), communicates with the StudentService to apply business rules, and returns the appropriate HTTP responses or UI templates.

4. Implementation Details

4.1. The Model (Student.java)
The Student class serves as the JPA Entity. It uses @Entity to map it to a database table. The primary key (id) is auto-generated using @GeneratedValue(strategy = GenerationType.IDENTITY). The fields include name, email, and course.

4.2. The Repository (StudentRepository.java)
This interface extends JpaRepository<Student, Long>, providing built-in CRUD functionalities without writing SQL queries. We defined custom query methods such as existsByEmail(String email) to check for data duplication, and findByCourse(String course) to retrieve students by course.

4.3. The Service (StudentService.java)
The service layer acts as a bridge between the Controller and the Repository. It houses the core business logic.
Duplicate Validation: Before adding a student, addStudent() checks if the email already exists using the repository's existsByEmail() method. If it does, it throws a RuntimeException.

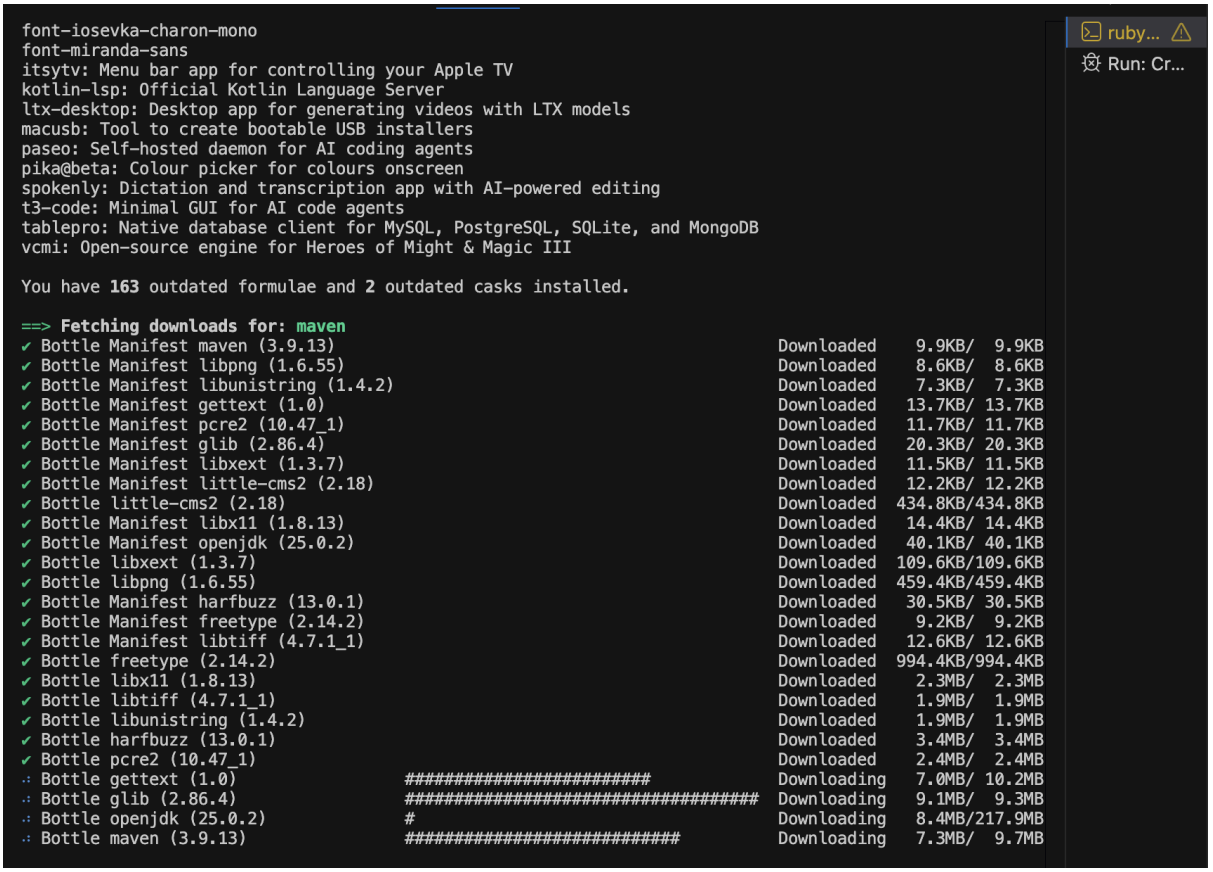
4.4. The Controller (StudentController.java)

- The controller serves both MVC web endpoints and REST API endpoints:
- GET / : Serves the main HTML interface with the student form and list.
- POST /add : Accepts form submissions to add a new student, using redirect attributes for success/error messages.
- GET /delete/{id} : Deletes a student by ID and redirects to the home page.
- POST /api/students : Accepts JSON payload and calls the service to save the student, returning a ResponseEntity.
- GET /api/students : Fetches all registered students as a JSON array.

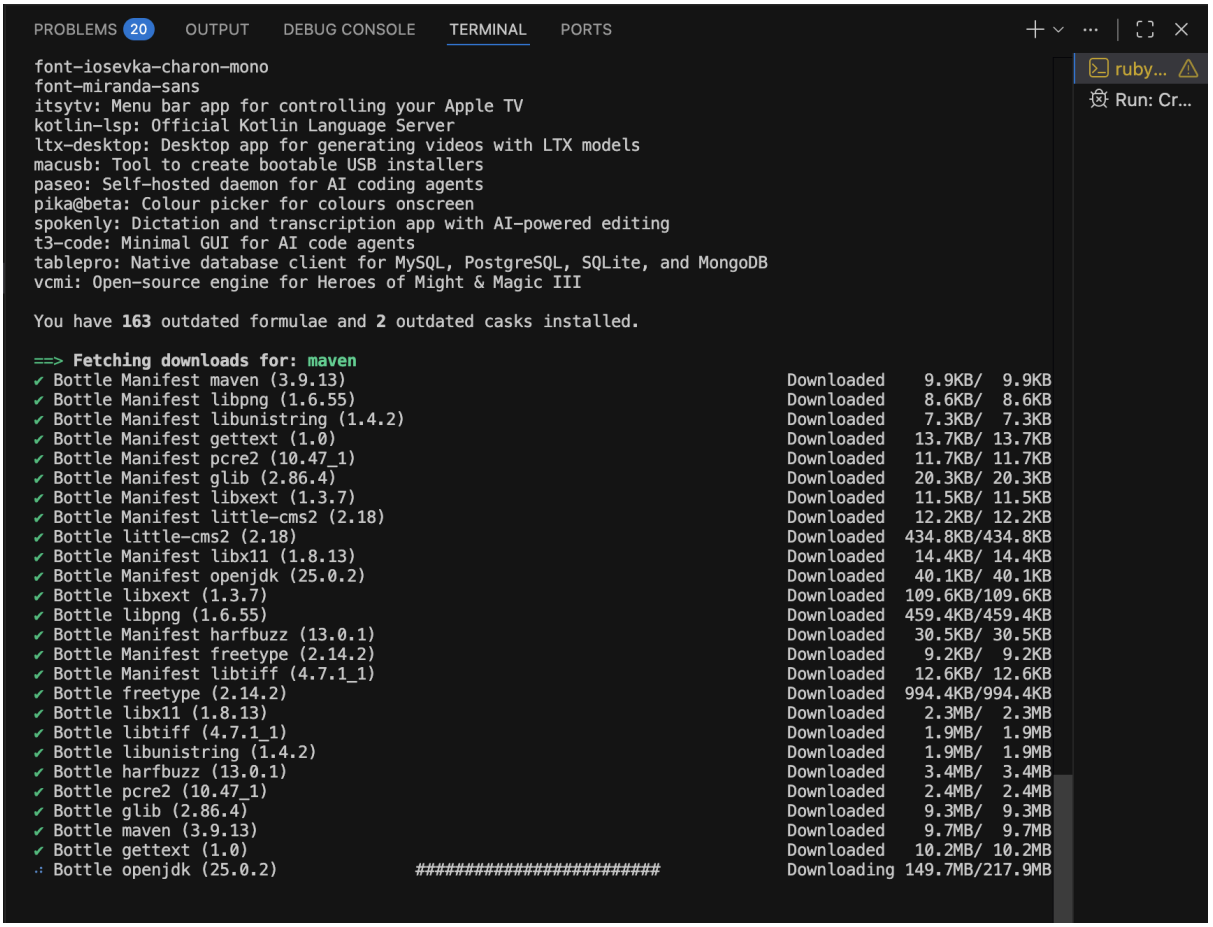
5. System Output & Verification (Screenshots)

5.0. Run the Project

The project was generated as a Maven-based Spring Boot application with the required dependencies: Spring Web, Spring Data JPA, H2 Database, and Thymeleaf, declared in pom.xml.



The H2 in-memory database is configured with spring.datasource.url=jdbc:h2:mem:studentdb. Hibernate's ddl-auto=updatesetting allows automatic table creation from entity definitions at startup. SQL logging is enabled for debugging purposes, and the H2 web console is made accessible at /h2-console.


PROBLEMS 20
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
ruby...
Run: Cr...
font-iosevka-charon-mono
font-miranda-sans
itsytv: Menu bar app for controlling your Apple TV
kotlin-lsp: Official Kotlin Language Server
ltx-desktop: Desktop app for generating videos with LTX models
macusb: Tool to create bootable USB installers
paseo: Self-hosted daemon for AI coding agents
pika@beta: Colour picker for colours onscreen
spokenly: Dictation and transcription app with AI-powered editing
t3-code: Minimal GUI for AI code agents
tablepro: Native database client for MySQL, PostgreSQL, SQLite, and MongoDB
vcmi: Open-source engine for Heroes of Might & Magic III
You have 163 outdated formulae and 2 outdated casks installed.
==> Fetching downloads for: maven
✔ Bottle Manifest maven (3.9.13) Downloaded 9.9KB/ 9.9KB
✔ Bottle Manifest libpng (1.6.55) Downloaded 8.6KB/ 8.6KB
✔ Bottle Manifest libunistring (1.4.2) Downloaded 7.3KB/ 7.3KB
✔ Bottle Manifest gettext (1.0) Downloaded 13.7KB/ 13.7KB
✔ Bottle Manifest pcre2 (10.47_1) Downloaded 11.7KB/ 11.7KB
✔ Bottle Manifest glib (2.86.4) Downloaded 20.3KB/ 20.3KB
✔ Bottle Manifest libxext (1.3.7) Downloaded 11.5KB/ 11.5KB
✔ Bottle Manifest little-cms2 (2.18) Downloaded 12.2KB/ 12.2KB
✔ Bottle little-cms2 (2.18) Downloaded 434.8KB/434.8KB
✔ Bottle Manifest libx11 (1.8.13) Downloaded 14.4KB/ 14.4KB
✔ Bottle Manifest openjdk (25.0.2) Downloaded 40.1KB/ 40.1KB
✔ Bottle libxext (1.3.7) Downloaded 109.6KB/109.6KB
✔ Bottle libpng (1.6.55) Downloaded 459.4KB/459.4KB
✔ Bottle Manifest harfbuzz (13.0.1) Downloaded 30.5KB/ 30.5KB
✔ Bottle Manifest freetype (2.14.2) Downloaded 9.2KB/ 9.2KB
✔ Bottle Manifest libtiff (4.7.1_1) Downloaded 12.6KB/ 12.6KB
✔ Bottle freetype (2.14.2) Downloaded 994.4KB/994.4KB
✔ Bottle libx11 (1.8.13) Downloaded 2.3MB/ 2.3MB
✔ Bottle libtiff (4.7.1_1) Downloaded 1.9MB/ 1.9MB
✔ Bottle libunistring (1.4.2) Downloaded 1.9MB/ 1.9MB
✔ Bottle harfbuzz (13.0.1) Downloaded 3.4MB/ 3.4MB
✔ Bottle pcre2 (10.47_1) Downloaded 2.4MB/ 2.4MB
✔ Bottle glib (2.86.4) Downloaded 9.3MB/ 9.3MB
✔ Bottle maven (3.9.13) Downloaded 9.7MB/ 9.7MB
✔ Bottle gettext (1.0) Downloaded 10.2MB/ 10.2MB
Bottle openjdk (25.0.2) ########################## Downloading 149.7MB/217.9MB

```
pranavhemanth@Pranavs-MacBook-Pro-M3 OOAD-S6 %/usr/local/mysql/bin/mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 9.4.0 MySQL Community Server - GPL

Copyright (c) 2000, 2025, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE studentdb;
Query OK, 1 row affected (0.006 sec)

mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| airlines_db        |
| assignment_sept4   |
| college            |
| company            |
| Fest_Database      |
| fest_management    |
| geocab             |
| information_schema |
| mysql              |
| performance_schema |
| shortly            |
| shortly_db         |
| studentdb          |
| sys                |
| university         |
+--------------------+
15 rows in set (0.013 sec)
```

```
pranavhemanth@Pranavs-MacBook-Pro-M3 OOAD-S6 %/usr/local/mysql/bin/mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 9.4.0 MySQL Community Server - GPL

Copyright (c) 2000, 2025, Oracle and/or its affiliates.

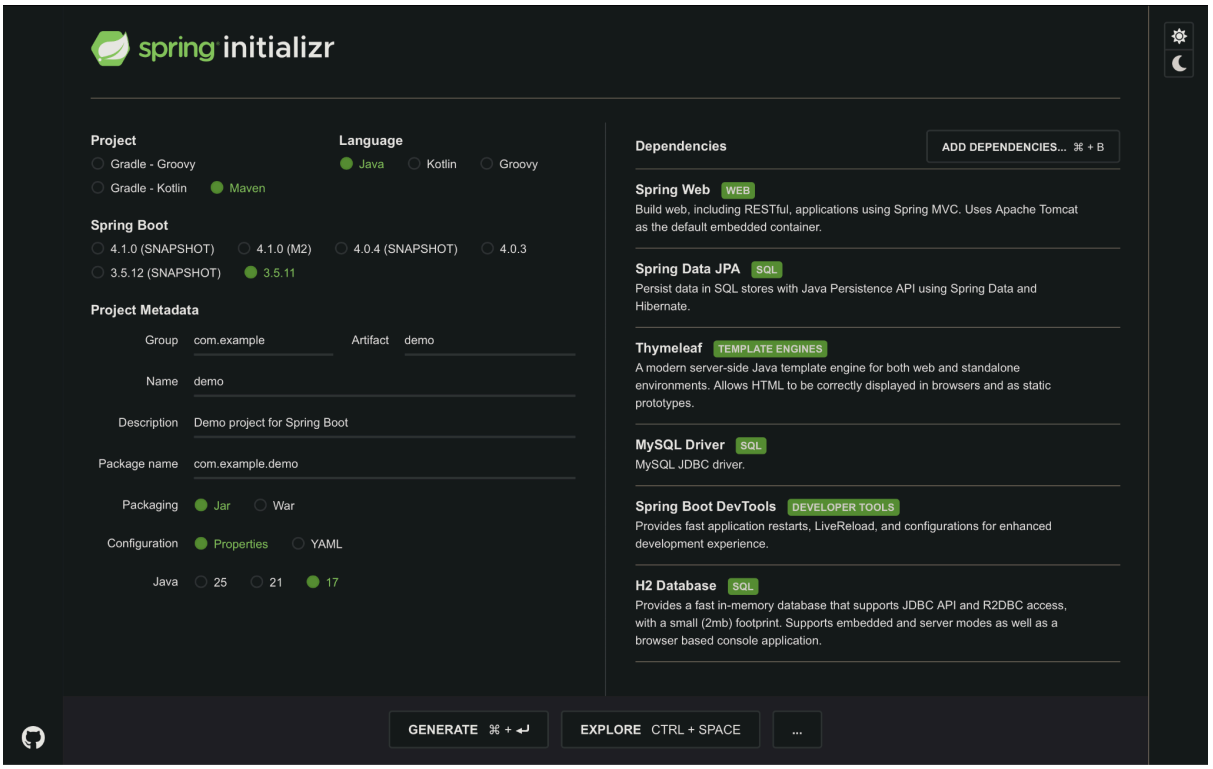
Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE studentdb;
Query OK, 1 row affected (0.006 sec)

mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| airlines_db        |
| assignment_sept4   |
| college            |
| company            |
| Fest_Database      |
| fest_management    |
| geocab             |
| information_schema |
| mysql              |
| performance_schema |
| shortly            |
| shortly_db         |
| studentdb          |
| sys                |
| university         |
+--------------------+
15 rows in set (0.013 sec)

mysql> exit
Bye
pranavhemanth@Pranavs-MacBook-Pro-M3 OOAD-S6 %
```

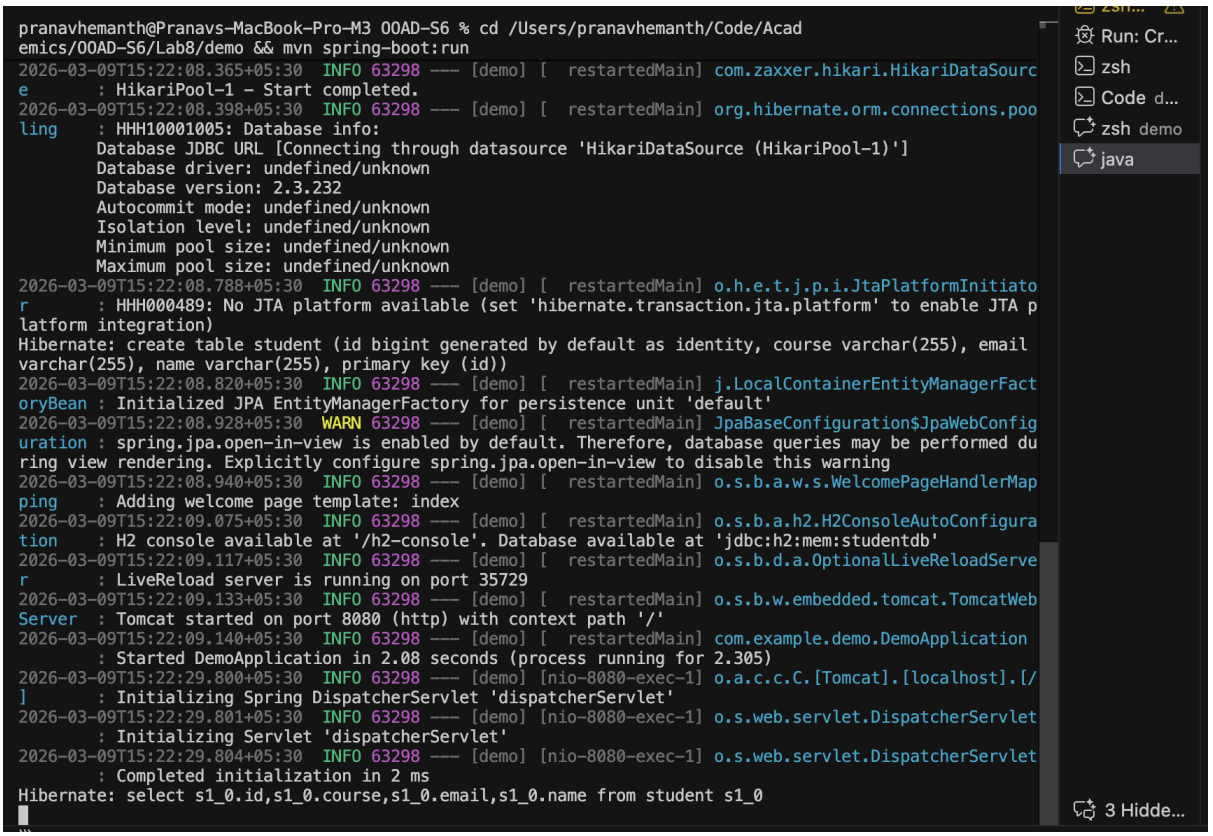


```
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab8 %ls
demo            demo.zip        Lab_Guide.pdf   MVC-Problem.pdf
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab8 %cd demo
pranavhemanth@Pranavs-MacBook-Pro-M3 demo %ls
HELP.md         mvnw            mvnw.cmd        pom.xml         src
pranavhemanth@Pranavs-MacBook-Pro-M3 demo %
```

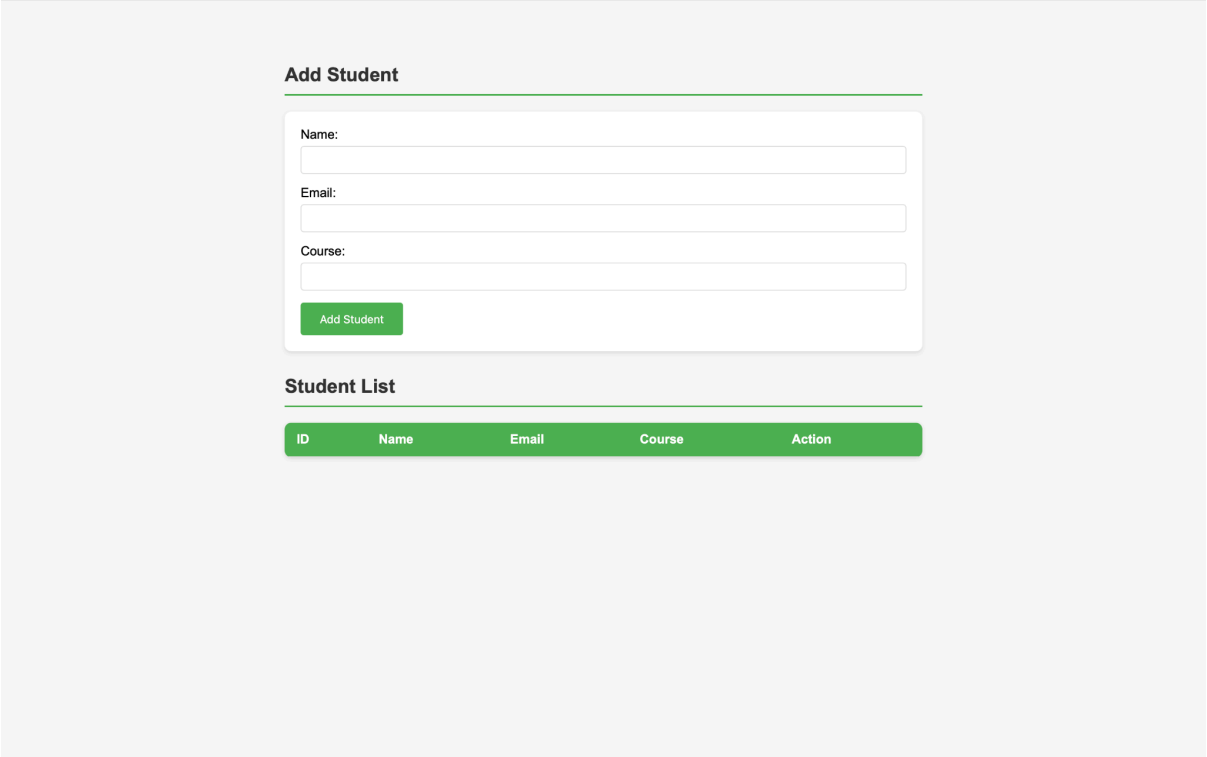
```
        : HHH000412: Hibernate ORM core version 6.6.42.Final
2026-03-09T15:20:46.675+05:30  INFO 61854 --- [demo] [  restartedMain] o.h.c.internal.RegionFactoryIniti
ator    : HHH000026: Second-level cache disabled
2026-03-09T15:20:46.829+05:30  INFO 61854 --- [demo] [  restartedMain] o.s.o.j.p.SpringPersistenceUnitIn
fo      : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-03-09T15:20:46.845+05:30  INFO 61854 --- [demo] [  restartedMain] com.zaxxer.hikari.HikariDataSourc
e       : HikariPool-1 - Starting...
2026-03-09T15:20:46.961+05:30  INFO 61854 --- [demo] [  restartedMain] com.zaxxer.hikari.pool.HikariPool
        : HikariPool-1 - Added connection conn0: url=jdbc:h2:mem:studentdb user=SA
2026-03-09T15:20:46.962+05:30  INFO 61854 --- [demo] [  restartedMain] com.zaxxer.hikari.HikariDataSourc
e       : HikariPool-1 - Start completed.
2026-03-09T15:20:46.997+05:30  INFO 61854 --- [demo] [  restartedMain] org.hibernate.orm.connections.poo
ling    : HHH10001005: Database info:
        Database JDBC URL [Connecting through datasource 'HikariDataSource (HikariPool-1)']
        Database driver: undefined/unknown
        Database version: 2.3.232
        Autocommit mode: undefined/unknown
        Isolation level: undefined/unknown
        Minimum pool size: undefined/unknown
        Maximum pool size: undefined/unknown
2026-03-09T15:20:47.423+05:30  INFO 61854 --- [demo] [  restartedMain] o.h.e.t.j.p.i.JtaPlatformInitiato
r       : HHH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA p
latform integration)
Hibernate: create table student (id bigint generated by default as identity, course varchar(255), email
varchar(255), name varchar(255), primary key (id))
2026-03-09T15:20:47.459+05:30  INFO 61854 --- [demo] [  restartedMain] j.LocalContainerEntityManagerFact
oryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-03-09T15:20:47.576+05:30  WARN 61854 --- [demo] [  restartedMain] JpaBaseConfiguration$JpaWebConfig
uration : spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed du
ring view rendering. Explicitly configure spring.jpa.open-in-view to disable this warning
2026-03-09T15:20:47.588+05:30  INFO 61854 --- [demo] [  restartedMain] o.s.b.a.w.s.WelcomePageHandlerMap
ping    : Adding welcome page template: index
2026-03-09T15:20:47.730+05:30  INFO 61854 --- [demo] [  restartedMain] o.s.b.a.h2.H2ConsoleAutoConfigura
tion    : H2 console available at '/h2-console'. Database available at 'jdbc:h2:mem:studentdb'
2026-03-09T15:20:47.769+05:30  INFO 61854 --- [demo] [  restartedMain] o.s.b.d.a.OptionalLiveReloadServe
r       : LiveReload server is running on port 35729
2026-03-09T15:20:47.788+05:30  INFO 61854 --- [demo] [  restartedMain] o.s.b.w.embedded.tomcat.TomcatWeb
Server  : Tomcat started on port 8080 (http) with context path '/'
2026-03-09T15:20:47.796+05:30  INFO 61854 --- [demo] [  restartedMain] com.example.demo.DemoApplication
        : Started DemoApplication in 2.184 seconds (process running for 2.435)
```

Jan - May 2026 Assignment SUBMISSION_UE23CS352B

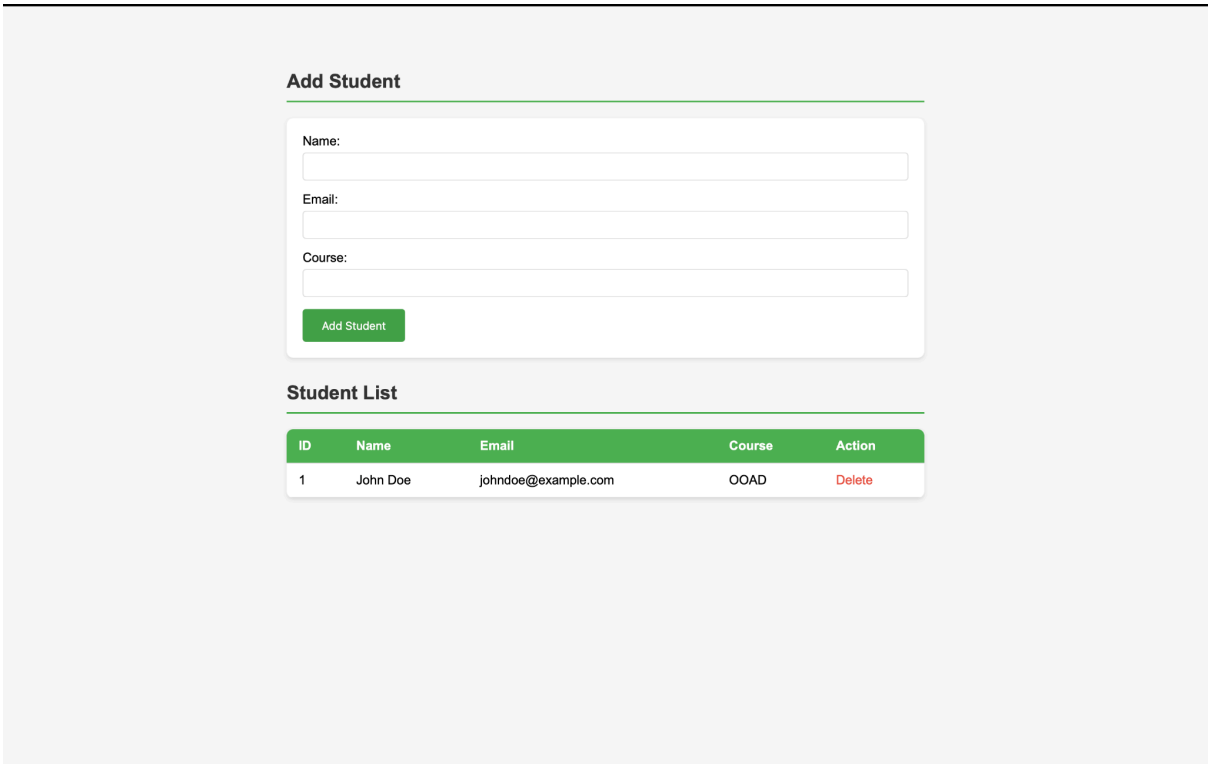




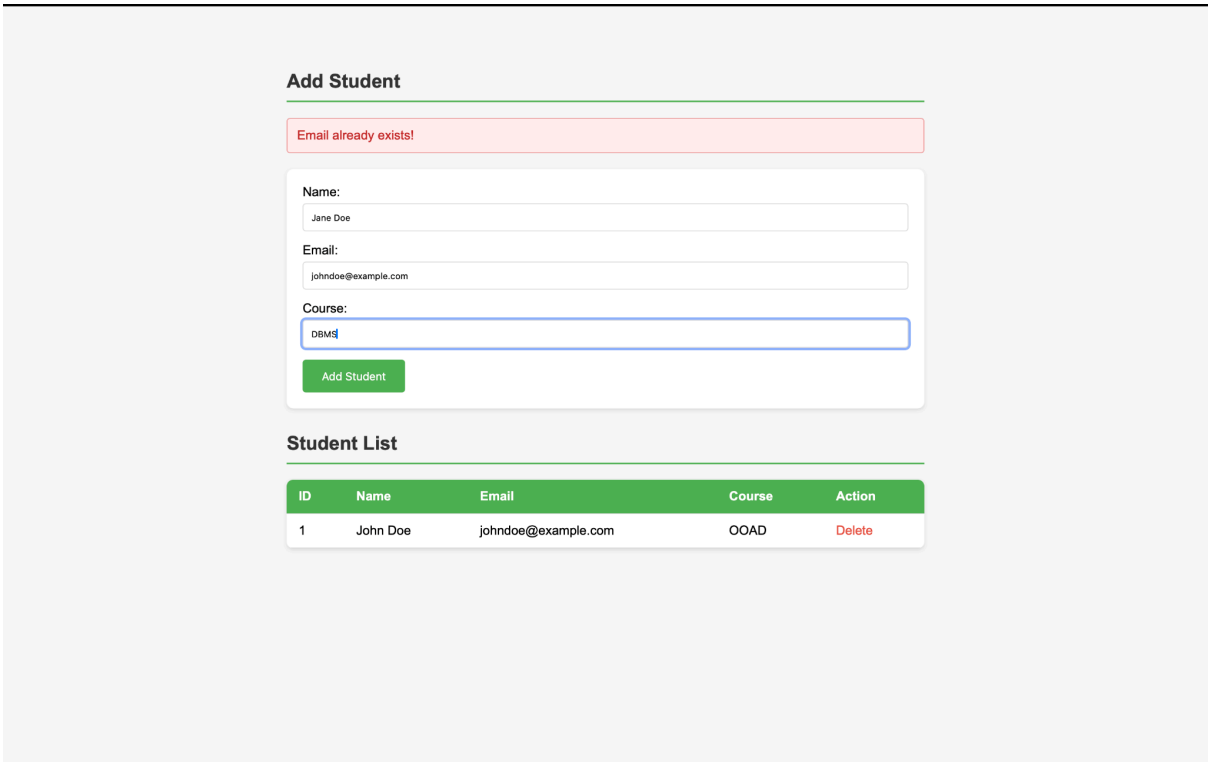
5.1. Main Interface (Empty State)
(The initial view of the registration form and empty table)

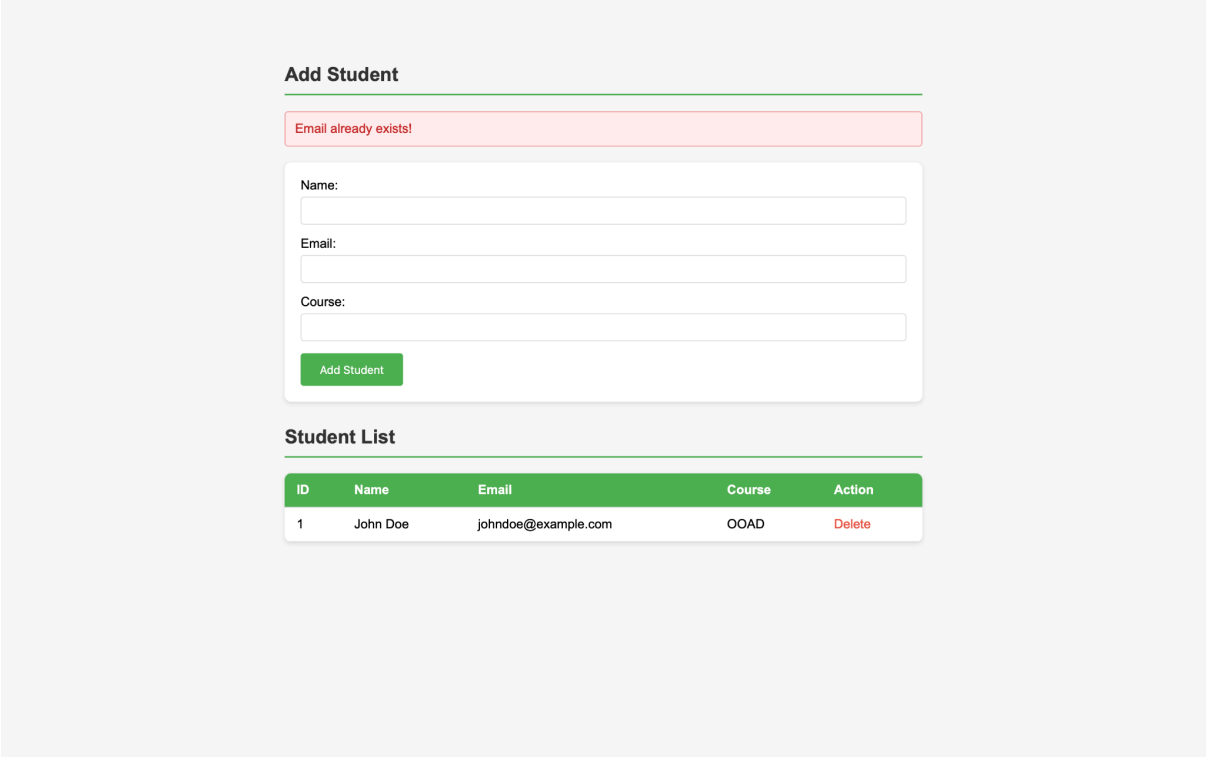


5.2. Successful Registration
(Demonstrating data insertion via the form)

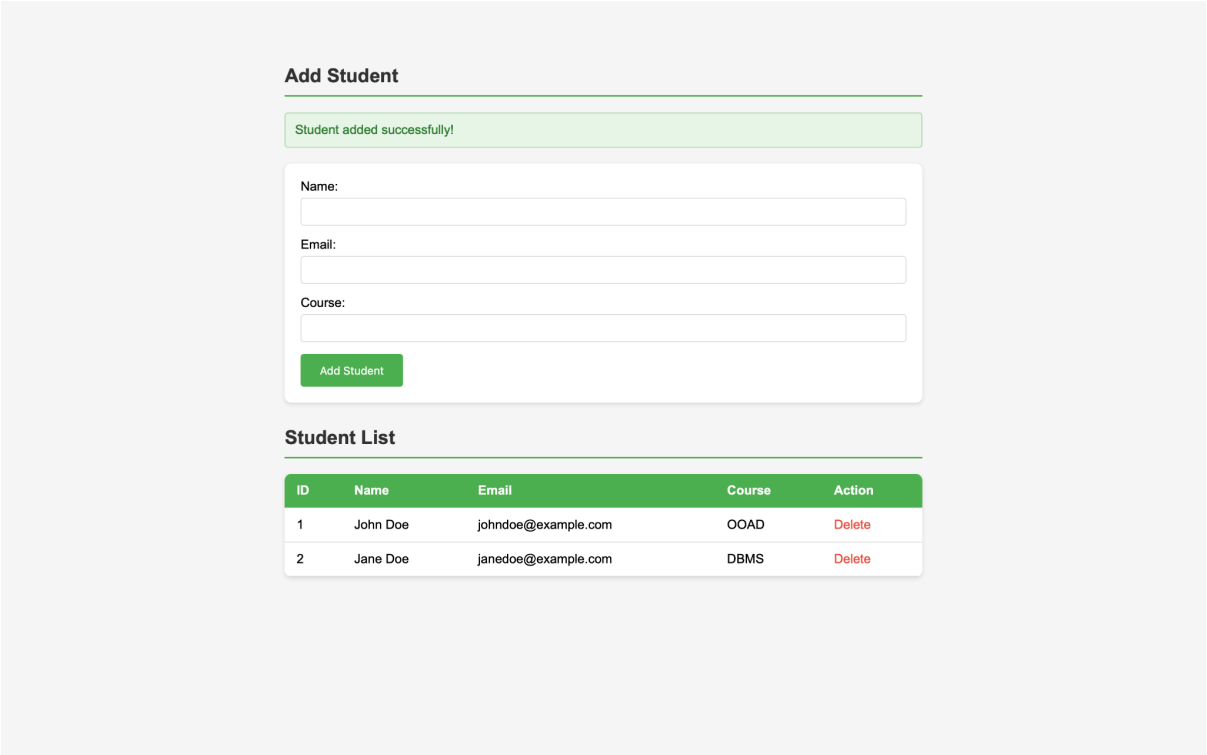


5.4. Business Logic: Duplicate Email Prevention
(Demonstrating the service layer rejecting a duplicate entry)





5.5. Business Logic: Non - Duplicate Email acceptance



5.6. Manipulating the database through http requests

